Lianjie (Marc) Shi

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EDUCATION & QUALIFICATIONS

University of British Columbia (UBC) — **Vancouver, Canada**

*Bachelor of Arts, Double Major in Statistics and Psychology (**Cumulative GPA: 92%; WES: 3.9/4.0**)* — *09/2020 – 05/2026*

- **Relevant Courses:**

Adv Statistical Inference I (STAT460/560)	95% (A+)	Adv Statistical Inference II (STAT461/561)	94% (A+)
Bayesian Statistics (STAT405)	93% (A+)	Design and Analysis of Experiments (STAT404)	94% (A+)
Time Series & Forecasting (STAT443)	91% (A+)	Real Analysis I (MATH320)	82% (A-)
Stochastic Processes (MATH303)	96% (A+)	Linear Regression (STAT306)	90% (A+)
Intro Statistical Inference (STAT305)	98% (A+)	Intro Probability (STAT302)	99% (A+)
Matrix Algebra (MATH221)	93% (A+)	Differential Equations (MATH215)	100% (A+)
Calculus III (MATH200)	92% (A+)	Calculus IV (MATH317)	93% (A+)
Calculus I (MATH104)	100% (A+)	Calculus II (MATH105)	93% (A+)

- **Honours & Awards:**
  - Dean's List (2023, 2024) – at least 27 credits with grades above 85%;
  - Work Learn International Undergraduate Research Award (WLIURA) – **$8,700** research award (2024);
  - Faculty of Arts International Student Scholarship – **$5,500** merit award (2024);
  - Dr. John and Barbara Petkau Scholarship – **$5,650** merit award for the fourth-year stats student with the highest standing (2025).

Peking University (PKU) — **Beijing, China**

Visiting Student (Non-degree) — *09/2026 – 01/2027*

- **Relevant Courses (in progress):** Deep Learning and AI; Machine Learning and AI

PUBLICATIONS & CONFERENCES

- Ou, C., Lou, N. M., Maheshka, C., **Shi, M**., Takemura, K., Cheung, B., & Heine, S. J. (2025). *Healthy sleep durations appear to vary across cultures. Proceedings of the National Academy of Sciences*, 122(19), e2419269122. https://doi.org/10.1073/pnas.2419269122
- Zhang, X., Wu, H., **Shi, L.**, & Savalei, V. (in revision). *Model identification made clear: From conceptual foundations to CFA identification rules.* Revise and resubmit at *Psychological Methods*. Preprint: https://osf.io/preprints/psyarxiv/pqa6y
- Zhang, X., Wu, H., Savalei, V., & **Shi, L.** (2026, July). *Model identification in CFA* [Symposium presentation]. In International Meeting of the Psychometric Society (IMPS), Seoul, South Korea.

RESEARCH EXPERIENCE

Directed Studies Project – Cross-Cultural Variation in Healthy Sleep Duration and Chronotype: A Multilevel Analysis

Supervisor: Dr. Steven Heine (Culture and Self Lab, UBC) — *09/2022 – 05/2024*

- Fit **mixed-effects (multilevel) models** in R to large-scale cross-national survey data, separating within- and between-country effects of chronotype and sleep duration on health outcomes.
- Estimated **cross-level interactions** between country-level sleep norms and individual sleep duration in predicting health.
- Wrote up the directed-studies project [paper]; contributed as the **4th author** on the published article in *PNAS* (2025).

Directed Studies Project – Selecting Confirmatory Factor Models for Balanced Likert Scales

Supervisors: Dr. Victoria Savalei (SEM Lab, UBC) & Dr. Xijuan (Cathy) Zhang (York University) — *09/2023 – 05/2024*

- Compared competing **CFA models** for balanced Likert scales, testing **nesting and equivalence relations** with the NET procedure in R.
- Validated the selected models on empirical data with **structural equation modeling** and model-fit assessment.
- Wrote up the directed-studies [paper]; gave an oral talk at the UBC Psychology Undergraduate Research Conference (2024).

Research Project – Sensitivity of $RMSEA_D$ to Baseline-Model Misspecification in Multi-Group Factor Models

Supervisor: Dr. Victoria Savalei (SEM Lab, UBC) — *05/2024 – 08/2025*

- Studied how **baseline-model misspecification** affects tests of **cross-group parameter equality** (measurement invariance) in multi-group factor models, using $RMSEA_D$ (a fit index for nested-model comparison) as the outcome.
- Developed an **R Shiny application** to visualize $RMSEA_D$ across **one- and two-factor models** under user-specified misspecifications (residual covariances, cross-loadings), supporting model comparison and residual analysis.

Research Project – Model Identification Rules for Confirmatory Factor Analysis

Supervisor: Dr. Xijuan (Cathy) Zhang (Quantitative Psychology Lab, York University) — *09/2024 – 05/2026*

- Evaluated **identification rules** for factor models with correlated errors and cross-loadings, applying **Wald rank tests** to check identifiability on simulated and empirical data.
- Wrote and ran **SEM code for every model in the paper's tables** on an empirical dataset (Rosenberg Self-Esteem Scale), building the paper's **empirical example** that verifies each identification rule on real data.
- **Third author** on the resulting methods paper (revise and resubmit, *Psychological Methods*); co-author on the IMPS 2026 talk.

SELECTED PROJECTS

More projects, code and reports: marclianjie.github.io

Bayesian Latent-State Models for Emotion Dynamics | *STAT 405 Course Project, UBC* *03/2026 – 04/2026*

- Built three **Bayesian latent-state models** of increasing complexity in **Stan** (ordered-logistic measurement, latent autoregression, lagged emotions) for experience-sampling data from 105 participants.
- Compared **NUTS with mean-field variational inference** via **PSIS-LOO**: latent time structure raised ELPD by about 1,900, and VI fell furthest behind HMC once temporal dependence entered [code].

Ablation of Batch Normalization and Data Augmentation in CNNs | *Deep Learning and AI, PKU* *10/2026*

- Ran a **controlled 2 × 2 ablation** (batch normalization × data augmentation) of a 320K-parameter CNN in **PyTorch** on 25,000 images, holding the architecture, seed, batch order and optimizer fixed.
- Augmentation raised validation accuracy by 7.3 points to 87.6% and removed the late-epoch decline; batch normalization mainly stabilized training, and the two effects did not stack [code].

Leakage-Free Model Comparison for Consumer Brand Choice | *Machine Learning and AI, PKU* *10/2026*

- Compared four tuned classifiers against a majority baseline in **leakage-free scikit-learn pipelines** (preprocessing fit inside each cross-validation fold); showed that 9 of 17 predictors were redundant.
- Locked the final model before opening the test set: 84.1% test accuracy and 0.92 AUC vs. a 61% baseline, holding at 0.89 AUC on a **temporal hold-out** of later weeks [code].

TECHNICAL PROFICIENCIES

- **R** (Mixed-Effects (Hierarchical) Models, Latent Variable Models, Simulation Studies, R Shiny App Development)
- **Stan** (Hamiltonian Monte Carlo, Variational Inference, PSIS-LOO)
- **Python** (NumPy, pandas, scikit-learn, Matplotlib, PyTorch, TensorFlow/Keras)
- **Java** (Object-Oriented Programming, Data Structures & Algorithms); **C** (Systems Programming, Linux)
- **SQL** (MySQL; Relational Database Design)